MIL-P-46439C (PA)
20 August 1976
SUPERSEDING
MIL-P-46439B (MU)
20 July 1965

MILITARY SPECIFICATION

PRIMER, ELECTRIC, M86
LOADING, ASSEMBLING AND PACKING

This specification is approved for use by Picatinny Arsenal, Department of the Army and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers the loading, assembling and packing for one type of primer designated as Primer, Electric, M86.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116 - Preservation, Methods of
MIL-P-223 - Powder, Black
MIL-I-46434 - Igniter, Electric, M59, Loading, Assembling and Packing
MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC: 1390

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MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

DRAWINGS

ARMY

8847424 - Primer, Electric, M86
8848806 - Carton, Packing, Ammunition, for Primer, Electric, M86
8848807 - Box, Packing, Ammunition, for Primer, Electric, M86

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer).

3. REQUIREMENTS

3.1 Material.-Materials shall be in accordance with applicable drawings and specifications.

3.2 Primers.-The loaded primers shall comply with all requirements specified on Drawing (dwg.) 8847424 and with all requirements specified in applicable specifications.

3.3 Moisture content of black powder.-The moisture content of the black powder at the loading station at the time of loading shall not exceed 0.5 percent when tested as specified in 4.3.1.

3.4 Electrical continuity.-The primers shall have an electrical resistance of not less than 0.80 ohm nor greater than 1.80 ohms when tested as specified in 4.3.2.

3.5 Lacquer in flash holes.-The lacquer in primer flash holes shall comply with illustration 1, 2 or 3 of Figure 1.

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3.6 Proving ground.-The primer, shall comply with the following requirements when tested as specified in 4.3.3.

a. The primer shall not fail to fire.
b. The primer body shall not rupture or become detached.
c. There shall be no audible hangfires.
d. The primer shall not squib.

3.7 Workmanship.-Loading and assembling shall be accomplished in a thorough, workmanlike manner. Each component and assembly shall be free of dirt, cracks, grease, rust, corrosion products or other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. Surface coatings shall be continuous except for a few slight scratches not exposing base material. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection and Standard Quality Assurance Provisions.-Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Quality Conformance Inspection

4.2.1 Inspection lot formation.-Inspection lots shall comply with the lot formation provision of MIL-A-48078. In addition, inspection lots of primers M86 shall contain:

a. Metal parts or metal parts assemblies from lots bearing the same lot interfix number from one manufacturer.
b. Black powder from not more than one lot.
c. Electric igniters from lots bearing the same lot interfix number from one manufacturer.

4.2.2 Examination -See MIL-A-48078

a. Sampling Plans. Unless otherwise specified in the Classification of Defects and Test Tables, sampling plans for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II. Continuous sampling plans in accordance with Standard MIL-STD-1235 may be used if approved by the procuring activity.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46439C (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.2.2.1	Primer assembly, prior to loading		8847424
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical 1	Any flash hole missing		100%	3.2	Gage
Major 101	Component damaged		.40%	3.2	Visual
102	Evidence of poor workmanship		.40%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46439C (PA)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.2.2.2	Charge, prior to loading	1	1	8847424
				NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Weight of charge	4.2.3.1	.40%	3.2	Balance
102	Moisture			3.3	4.3.1
Minor					
201	Foreign matter in charge		.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46439C (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.2.2.3	Primer assembly prior to assembly of diaphragm				8847424
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Charge missing		.40%	3.2	Visual
102	Liner improperly positioned		.40%	3.2	Visual
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46439C (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.2.2.4	Primer assembly		8847424
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Stake missing, or inadequate		.40%	3.2	Visual
102	Flash hole sealing improper		.40%	3.5	Visual
103	Split or cracked head		.40%	3.2	Visual
104	Igniter above flush with surface of head		.40%	3.2	Visual
105	Depth to diaphragm (see Note)		.40%	3.2	Gage
106	Outer diaphragm improperly assembled		.40%	3.2	Visual
107	Diaphragm or liner missing, punctured or improperly sealed		.40%	3.2	Visual
108	Electric continuity	4.2.3.2		3.4	4.3.2
109	Functioning	4.2.3.3		3.6	4.3.3
Minor					
201	Total length, maximum (max.)		.65%	3.2	Gage
202	Threads damaged		.65%	3.2	Visual
203	Marking misleading or unidentifiable		.65%	3.2	Visual
204	Evidence of poor workmanship		.65%	3.7	Visual

NOTES: This depth pertains to inner diaphragm and inspection shall be performed prior to assembling end diaphragm.

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46439C (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.2.2.5	Carton, prior to sealing		8848806
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Any packing material missing		.40%	3.2	Visual
102	Separator damaged to the extent that primers touch each other		.40%	3.2	Visual
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46439C (PA)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.2.2.6	Carton, sealed				8848806
					NEXT HIGHER ASSEMBLY
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major	None defined				
Minor					
201	Sealing strip torn, badly wrinkled, incomplete or otherwise fails to seal carton completely		.65%	3.2	Visual
202	Carton torn, cut or punctured		.65%	3.2	Visual
203	Contents loose to an extent greater than permitted by packing design		.65%	3.2	Manual
204	Marking missing, misleading or unidentifiable		.65%	3.2	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46439C (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.2.2.7	Sealed barrier bag		8848806
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Bag, punctured, torn or cut		.65%	3.2	Visual
102	Seal mislocated or incomplete		.65%	3.2	Visual
Minor					
201	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

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.QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46439C (PA)

PARAGRAPH	TITLE	SHEET	DRAWING NUMBER
4.2.2.8	Box, Packing prior to sealing	1 OF 1	8848807
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major 101	Carton missing		.40%	3.2	Visual
Minor	None defined				

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-46439C (PA)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.2.2.9	Box Packing, sealed		8848807
			NEXT HIGHER ASSEMBLY

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Protruding nail		.40%	3.2	Visual
102	Board broken or loose		.40%	3.2	Visual
103	Strapping missing, broken or loose		.40%	3.2	Visual/Manual
Minor					
201	Contents loose		.65%	3.2	Manual
202	Strapping improperly engaged or mislocated		.65%	3.2	Visual
203	Marking missing, misleading or unidentifiable		.65%	3.2	Visual
204	Evidence of poor workmanship		.65%	3.7	Visual

NOTES:

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4.2.3 Testing

4.2.3.1 Moisture content of black powder (see 3.3) Major defect.-The contractor shall provide adequate controls to assure the compliance of each lot with the applicable requirement. For verification, one sample of black powder shall be selected at the loading station, during each eight hours production. If the moisture content of the sample exceeds the applicable requirement the primers loaded since the last acceptable sample shall be rejected. The test shall be performed as specified in 4.3.1.

4.2.3.2 Electrical continuity (see 3.4) Major defect.- This test shall be performed 100 percent as specified in 4.3.2 using the applicable equipment in accordance with 4.2.4. Any unit exhibiting a resistance outside the permitted range shall be classed defective and removed from the lot.

4.2.3.3 Proving ground (see 3.6).-The primer assemblies shall be observed for evidence of the following defects:

Table I

Defects	Classification
Body ruptures or becomes detached	Critical
Failure to fire	Major
Assembly hangfires	Major
Assembly squibs	Major

4.2.3.3.1 Beginning with the first lot produced and continuing until three consecutive lots have complied with the acceptance criteria specified, 35 primers shall be randomly selected from each lot for this test. If a critical defect occurs or if two or more major defects occur, the lot shall be rejected.

4.2.3.3.2 After three consecutive lots have met the criteria specified in 4.2.3.3.1, 10 primers shall be randomly selected from each lot. If a critical defect occurs or if one or more major defects occur, the lot shall be rejected.

4.2.3.4 Check test for deterioration of Igniters, Electric M59, Major defect (see 6.4).-If the igniters have been stored under normal conditions more than 2 years or for any period of time under adverse conditions, however brief, since previous tests, the igniter lot shall be subjected to, and must comply with, the check test specified in Specification MIL-I-46434 immediately before assembly into the primers of this specification.

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4.2.3.5 Heat seal test of barrier bag (see dwg. 8848806) (Major defect).-Sampling, acceptance criteria and test procedures for this test shall be as prescribed in Specification MIL-P-116 for the "Heat Seal Test".

4.2.4 Inspection Equipment.-The inspection equipment required to perform the examinations and tests prescribed herein is described in the paragraph reference/inspection method column in the tables starting with paragraph 4.2.2.1. The contractor shall submit for approval inspection equipment designs in accordance with the terms of the contract. See section 6. of MIL-A-48078.

4.3 Test methods and procedures

4.3.1 Moisture content of black powder.-The moisture content of the black powder shall be tested in accordance with the procedure specified in Specification MIL-P-223.

4.3.2 Electrical continuity.-The resistance of the primer shall be taken between the electrode as one contact and the primer head as the other. The applied current shall be limited to 20 milliamperes, maximum. Any unit exhibiting a resistance outside the permitted range shall be classed defective.

4.3.3 Proving ground.-Testing shall be performed at a Government proving ground. The sample primers shall be assembled into an M393A1 cartridge and fired from a M68 gun tube. Observation shall be made for compliance with each requirement.

5. PREPARATION FOR DELIVERY

5.1 Packaging and Marking

5.1.1 Level C.-The primers shall be packaged and marked in accordance with dwg. 8848806.

5.2 Packing and marking.

5.2.1 Level C.-The packaged primers shall be packed and marked in accordance with dwg. 8848807.

6. NOTES

6.1 Ordering Data See MIL-A-48078.

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6.2 Submission of Designs for Approval See 6.2.3 of MIL-A-48078.-Submit equipment designs, as required, to Commander, Picatinny Arsenal, ATTN: SARPA-QA-T-T, Dover, New Jersey 07801.

6.3 Check test for possible deterioration of primers.-The following tests are designed to prevent loading major ammunition items with primers which have deteriorated. The tests should be performed on each lot of M86 Primers which have been stored for more than two years or which have been subjected to any adverse conditions, however brief, within that period of time. If the specification for the ammunition item into which the M86 Primer is assembled, does not require the performance of the following check test, the contracting officer shall contractually provide for compliance of check test requirements.

6.3.1 Moisture content of black powder.-A sample of 25 primers shall be selected for this test. The black powder shall be removed from the body by any suitable means. The moisture content of the powder shall be determined in accordance with the methods specified in Specification MIL-P-223. If the moisture content of the powder tested exceeds 0.7 percent, the lot of primers shall be rejected.

6.3.2 Electrical resistance.-This test shall be performed 100 percent. Any primer assembly which fails to comply with the requirement specified in 3.4 shall be classed defective and removed from the lot. The test shall be performed as specified in 4.3.2.

6.3.3 Functioning.-300 primers shall be selected from each lot for this test. The igniters (Igniter, Electric, M59) shall be removed from the primers and tested as specified in Specification MIL-I-46434. If three or more igniters fail to comply with the applicable requirement the lot shall be rejected. At the option of the Contractor, complete M86 primer assemblies can be tested as specified in Specification MIL-I-46434. Observation shall be made on functioning of the complete primer assembly. If three or more primers fail to comply with the applicable requirement the lot shall be rejected.

6.4 Check test reimbursement.-The contracting officer shall arrange to have the contractor reimbursed for the expense incurred in the performance of the Igniter, Electric, M59 check test.

Custodian:
Army- PA

Preparing Activity:
Army-PA

Project Number: 1390-A145

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ARTILLERY PRIMERS
STANDARDS FOR PAINT CONDITIONS IN FLASH HOLES

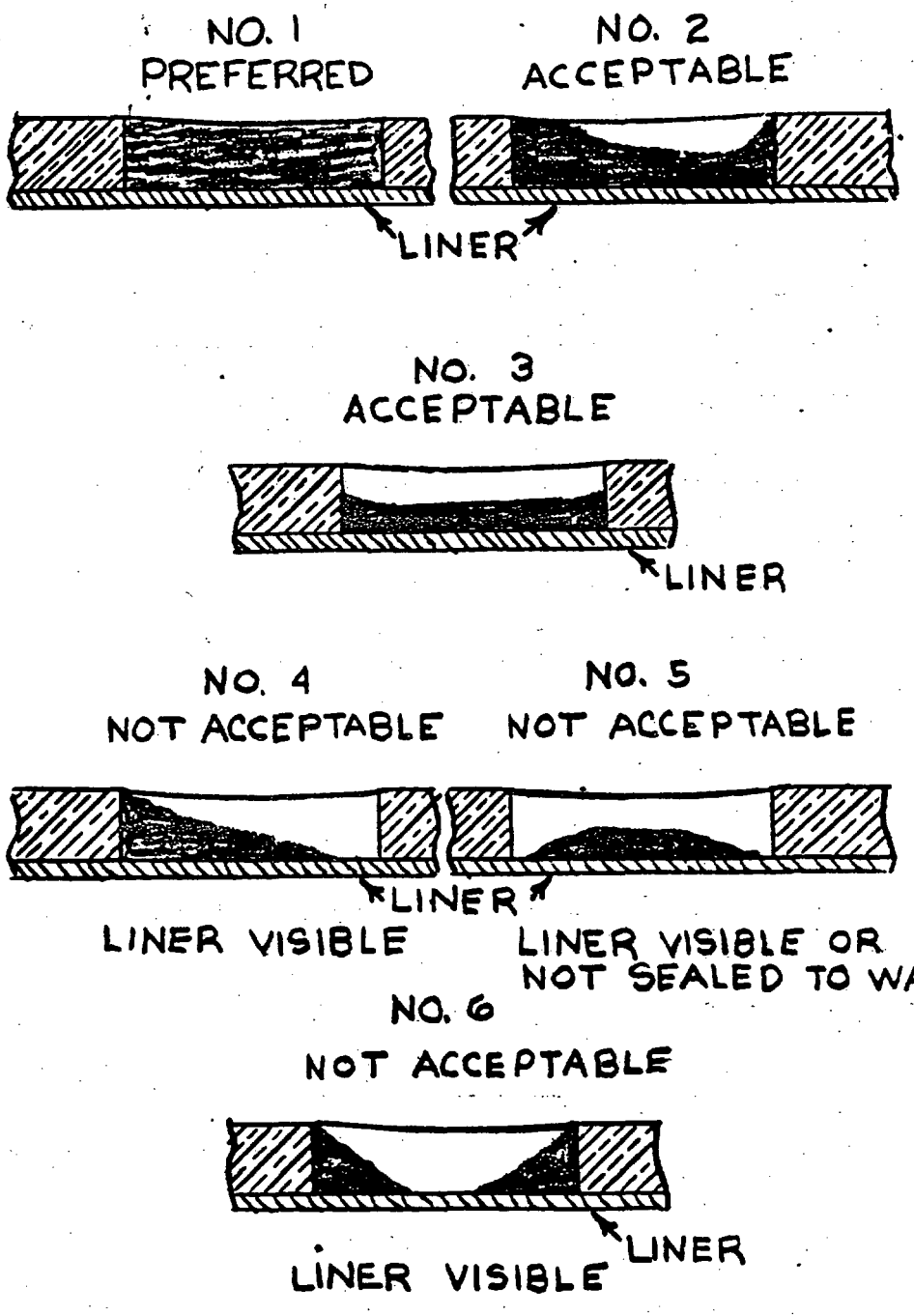


FIGURE 1

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OMB Approval
No. 22-R255

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DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRAC

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

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